

Kabeen Kim

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Research Interests: Knowledge Graphs · GraphRAG · Multi-hop Retrieval & Reranking · Grounded LLMs · Document Understanding

Education

Duksung Women's University

B.S. IN COMPUTER ENGINEERING & INFORMATION STATISTICS | GPA: 4.15 / 4.5

Seoul, South Korea

Mar 2021 – Feb 2027 (Expected)

University of Mississippi

EXCHANGE STUDENT, COMPUTER SCIENCE

Mississippi, USA

Aug 2024 – Dec 2024

Publications

- [1] Haopeng Chen, Yihao Ai, **Kabeen Kim**, et al. "UDAPose: Unsupervised Domain Adaptation for Low-Light Human Pose Estimation." *Computer Vision and Pattern Recognition (CVPR)*, 2026. [Paper] [Code]
- [2] **Kabeen Kim**[†], Yena Kim*, Minjeong Seo*. "ConFuse: Context-Aware Fusion of LLMs and GBDTs for Sleep Prediction from Lifelog Data." *International Conference on ICT Convergence (ICTC)*, 2026. (Poster) [Code] *Equal contribution. [†]Presenter.
- [3] **Kabeen Kim**[†], et al. "CoDU: Multi-Stage Approach for Complex Document Structure Understanding." *Korea Data Base Conference (KDBC)*, 2025. (Oral) [Code]
- [4] **Kabeen Kim**[†], et al. "SPTC (Single-Pass Tree Chain-of-Thought): An Efficient Prompting Scheme for Small Language Models." *Korea Computer Congress (KCC)*, 2025. (Poster) [Paper]

Research Experience

Electronics and Telecommunications Research Institute (ETRI)

Daejeon, South Korea

RESEARCH INTERN – KNOWLEDGE GRAPH & MULTI-HOP GRAPH RAG

Jul 2026 – Aug 2026

- Extracted OpenIE triples from technical manuals, verified them against source passages, and built a NetworkX graph and Qdrant index with **~38K semantic relations and ~366K edges**. Curated **250 multi-hop QA questions** with a Ph.D. researcher.
- Diagnosed bridge evidence loss, then revised beam search to keep prior paths and cumulative scores and **reranked connected paths with BGE**, raising full gold evidence coverage **from 62/250 to 202/250** and Recall@5 from 0.458 to 0.861.
- Increased LLM judge correctness **from 2.84 to 3.86 out of 5** by grounding Qwen2.5-32B in 7 evidence passages from the top 5 paths, outperforming HippoRAG, Microsoft GraphRAG, and KG²RAG in-domain; adopted in an ongoing lab project.

Data Mining & Knowledge Discovery (DMKD) Lab, Duksung Women's University

Seoul, South Korea

UNDERGRADUATE RESEARCHER – LLM REASONING, DOCUMENT AI & RAG

Mar 2025 – Feb 2026

- Led the development of **SPTC [4]** to generate, evaluate, and select in one prompt, attaining **59.7% semantic matching accuracy** (5 small LMs, 500 CommonsenseQA/OpenBookQA items), within 1.4 pp of Tree-of-Thought in ~33% less time.
- Owned **CoDU [3]** layout detection as team lead, raising **F1 (IoU ≥ 0.5) from 0.463 to 0.580** through DocLayout-YOLO tuning and box correction. The final team system achieved 0.617 F1 and 3rd place in the Samsung AI Challenge.
- Doubled product model identification **F1 from 0.3 to 0.6** on 100 queries as RAG lead, combining Pinecone retrieval, Cohere reranking, query normalization, and prompt tuning; led to a follow-up contract.

Vision and Multimodal Intelligence Lab, University of Mississippi

Mississippi, USA

UNDERGRADUATE RESEARCHER – HUMAN POSE ESTIMATION IN LOW LIGHT

Aug 2024 – Apr 2026

- Initiated the collaboration through direct faculty outreach and established a **taxonomy of 15 failure categories** for low-light pose estimation.
- Trained and benchmarked domain adaptation models for UDAPose [1] (**+10.1 AP** on ExLPose-test, +7.4 AP cross-dataset), continuing through biweekly remote meetings after returning to Korea.

Work Experience

Boston Consulting Group (BCG)

Seoul, South Korea

RESEARCH ANALYST INTERN

Mar 2026 – May 2026

- Fine-tuned Qwen models (4B–14B) with LoRA via Unsloth on 1,200 domain QA pairs, **raising LLM judge scores from 2.3 to 3.3** (out of 5). Gains appeared only in the 14B models, with Qwen2.5-14B-Instruct performing best.
- Automated QA data checks (deduplication, grounding) and an evaluation framework using EM, token F1, and LLM judge scores.
- Built a slide parser and a value-extraction agent, benchmarking 9 LLMs for RAG QA. The parser with Claude Sonnet 4.6 **ranked 1st by LLM judge score** among tested pairs.
- Integrated **30M+ rows of battery manufacturing process data** across tables into datasets for analysis using PostgreSQL, Python, Spotfire, and AWS EC2/S3.

Selected Projects

ConFuse: Context-Aware Fusion of LLMs and GBDTs for Lifelog Sleep Prediction

 github.com/kbannie/ConFuse

RESEARCH TEAM LEAD, TOBIG'S BIG DATA & AI SOCIETY

May 2026 – Sep 2026

- Spearheaded fusion of ML and LLMs for small lifelog datasets, designing inputs with 21 personal baseline features and trends over 3 days, tuning **Qwen2.5-7B with QLoRA (4-bit)**, and reducing log loss from 0.6782 to 0.6735 through calibration.
- Engineered fusion weights for each sample using input quality assessed by GPT-4.1, limiting LLM weights to 0–45% and achieving **0.6415 mean log loss**, 0.0115 below ML alone and 0.0030 below fixed 50:50 fusion [2].

Image Emotion Classification for a Multisensory Exhibition

 github.com/4CO4CO/4co4co-ai

AI LEAD, CAPSTONE PROJECT

Feb 2025 – Nov 2025

- Designed a **classifier for 17 emotions** (Moondream2 captions → CLIP image and text features → MLP), using YOLO to route face/behavior and scene analysis, including landscapes without people.
- Expanded 12,103 person samples with ~6,000 captioned natural scenes, achieving **Macro-F1 0.416 / Micro-F1 0.592**. Exhibited the system for 3 days with 95% visitor satisfaction.

Leadership

Duksung Global Challenger

TEAM LEAD & PROJECT PROPOSER, “BE A CONNECTOR” (WeCON)

Mar 2025 – Sep 2025

- Secured **KRW 10 million in university funding** to develop a Bluetooth Low Energy (BLE) beacon app connecting the campus with local businesses.
- Investigated BLE/UWB tradeoffs and user needs through English interviews with experts in four European countries, translating them into app features, beacon placement, and a budget, earning an **Encouragement Award**.

Teaching & Volunteering

Chungnam National University & NAQS (Korean Government Agency)

TEACHING ASSISTANT, N8N & RAG WORKSHOP

Sep 2025 – Oct 2025

- Authored **170 slides** and delivered live demos on API authentication, Docker, and n8n for a professor's special lectures (20 hours, 50 participants).
- Prototyped a RAG Q&A lab (national statistics & Google Drive → OpenAI embeddings → Pinecone → n8n/GPT-4o-mini) and **3 workflow automation labs** for public sector staff.

CJ OliveNetworks SW Creative Camp

UNIVERSITY VOLUNTEER INSTRUCTOR, CJ UNIT (8TH COHORT)

Mar 2023 – Jun 2023

- Taught **40 hours of LEGO robotics and Python coding** to middle school students in Jeju through remote classes (3 hours/week).
- Co-designed an ESG curriculum with 3 sessions on wave, solar, and wind energy, including LEGO assembly guides, sensor and motor control algorithms, and Python examples.

Honors & Awards

2025 **3rd Place (out of 210 teams)**, Samsung AI Challenge – Visually-Rich Document Understanding

2025 **Encouragement Award**, DS Regional Collaboration Idea Competition

2024 **31st Mirae Asset Overseas Exchange Scholarship**, Mirae Asset Park Hyeon Joo Foundation

2024 **Excellence Award**, TAVE 13th Data Analysis Project

2024 **2nd Prize**, DS Global Vision Makers Innovation Idea Competition

2021–22 **Academic Excellence Scholarship (3 awards)**, Duksung Women's University

Skills & Languages

- **Knowledge Graph & Retrieval**: GraphRAG, multi-hop retrieval, evidence reranking, graph–vector search, NetworkX, Qdrant
- **LLM & RAG**: LLM fine-tuning, LoRA/QLoRA (Unsloth), RAG, prompt engineering, LLM judge evaluation, LangChain, Pinecone
- **Statistics & Machine Learning**: regression, time series and multivariate analysis, data preprocessing, visualization, ensemble learning, XGBoost, LightGBM
- **Deep Learning & Computer Vision**: PyTorch, OpenCV, YOLO, CLIP
- **Programming & Systems**: Python, R, C++, SQL (PostgreSQL), AWS EC2/S3, Docker, Git, Weights & Biases, n8n
- **Certifications & Languages**: SQLD (SQL Developer), English (IELTS 7.0, OPIc AL), Korean (Native)